

Shantanu Sharma

AI/ML Engineer | GenAI Systems | Backend Infrastructure

Haryana, India | Open to Remote (Worldwide) | [linkedin.com/in/mathoholic](https://www.linkedin.com/in/mathoholic) | mathoholic.com | github.com/Mathoholic

AI/ML Engineer with hands-on experience building production-grade GenAI systems, event-driven SaaS platforms, and scalable backend infrastructure. Served as tech lead for a 12-person cross-functional engineering team, owning architecture decisions, tech stack choices, PR reviews, and end-to-end feature delivery. Built and shipped a SaaS automation platform actively used by 50+ businesses processing thousands of campaigns monthly. Strong across the full stack: backend, frontend, and mobile. Open to remote roles worldwide.

Technical Skills

AI/ML & GenAI: LangChain, RAG Pipelines, ChromaDB, Vector Databases, Embeddings, NLP, Recommendation Systems, Prompt Engineering, OpenAI API

Languages: Python, Go, SQL, TypeScript

Backend: FastAPI, Django, Flask, Gin, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Snowflake, Redis

Cloud & DevOps: AWS (EC2, RDS, S3, ECS, Chime, EventBridge, SQS), Docker, CI/CD, Azure DevOps

Messaging: RabbitMQ, Celery, Dead-Letter Queuing

Integrations: Third-Party API Integrations, Webhooks, Payment Gateway Integrations, VoIP Implementation

Data & Analytics: NumPy, Pandas, Airflow, Power BI, Tableau, dbt

Experience

AI/ML Engineer - Tech Lead – Centocode Technologies

Sep 2024 – Present / Remote

- Served as tech lead for a 12-person cross-functional team across backend, frontend, and mobile; responsible for tech stack decisions, system architecture, PR reviews, and sprint delivery
- Conducted technical research and feasibility scoping for new product features; translated business requirements into implementation roadmaps
- Built and maintained a SaaS automation platform serving 50+ active businesses, processing thousands of campaigns monthly: webhook ingestion, message workflows, rule-based chatbot, campaign execution engine on RabbitMQ (~150K+ messages/month) with retry logic and dead-letter queuing, and VoIP implementation for call workflows
- Integrated third-party APIs and services into a unified commerce automation platform, including webhook-based connectors, lead ingestion, and payment integrations
- Built production RAG systems using LLMs and ChromaDB, reducing information retrieval time by ~40% through optimized embedding and retrieval pipelines
- Developed natural language search over structured datasets, enabling non-technical users to query data without SQL
- Designed and deployed NLP-based recommendation engines integrated into a React Native mobile app
- Developed an LLM-based Interview Platform on AWS (Chime, EventBridge, SQS, RDS): real-time interview sessions with asynchronous AI evaluation pipelines
- Contributed across the full product stack: backend APIs (FastAPI/Go), frontend features, and React Native mobile advisory inputs

Senior Data Scientist – 1ViewApps

Feb 2024 – Aug 2024 / Remote

- Built AI-powered search, chatbot, and data visualization features using LLMs and LangChain
- Designed modular ETL pipelines for structured and semi-structured data from multiple sources
- Automated daily data workflows using Airflow, improving pipeline reliability and reducing manual intervention
- Implemented CI/CD pipelines across AWS and Azure DevOps

Associate Data Scientist – KDataScience Solutions Pvt. Ltd.

Jun 2023 – Feb 2024

- Built Python-based data pipelines for large-scale ingestion and processing
- Designed Power BI and Tableau integration pipelines using Python and MSSQL
- Developed backend services for on-demand data processing with focus on performance and scalability

Writing & Public Work

Technical Blog: mathoholic.com/blog

Writing on backend engineering, distributed systems, and AI/ML in production. Recent posts include RabbitMQ incident diagnosis, Docker build optimization, and reverse proxy architecture.

Public Projects: github.com/Mathoholic

Worker pool systems, concurrent job processors, RAG services, auth playground, and distributed backend infrastructure in Go and Python.

Certifications

- Python for Everybody Specialization – University of Michigan – Coursera, May 2023
- Visual Analytics with Tableau – UC Davis – Coursera, Apr 2023
- The Complete SQL Bootcamp: Zero to Hero – Udemy, May 2023
- Microsoft Power BI Desktop for Business Intelligence – Udemy, Dec 2023
- Data Engineering with dbt – LinkedIn Learning, Jan 2024

Education

B.A. Mathematics (Hons) – Dr. B.R. Ambedkar University Delhi

2015–2020